

Total No. of Questions : 4]

SEAT No. :

P5102

[Total No. of Pages : 2

[6187] 509

T.E. (Electronics & Computer Engineering) (Insem)
DATA COMMUNICATION
(2019 Pattern) (Semester-I) (310343)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4.
- 2) Figures to the right side indicate full marks.
- 3) Assume suitable data, if necessary.
- 4) Neat diagrams must be drawn wherever necessary.

- Q1)** a) Draw and explain OSI model in detail. [5]
b) Explain various data transmission modes with suitable example. [5]
c) Write a short on: [5]
i) Hub
ii) Switch
- Q2)** a) What are different types of guided and unguided transmission media? Explain any two types of guided media. [5]
b) Write a short on : [5]
i) Router
ii) Bridge
c) Explain unguided media and its types. [5]
- Q3)** a) Explain the different noise sources. [5]
b) Calculate thermal noise voltage for a bandwidth of 1MHz and value of resistance is [5]
i) 20 Ω
ii) 50 Ω
iii) 100 K Ω
iv) 200M Ω assume room temp of 27°C

P.T.O.

c) Explain the following terms: [5]

- i) Signal to Noise ratio
- ii) SNR in Tandem Connection

OR

Q4) a) Small signal amplifier operating at 10 to 12 MHz frequency range has a 200 K Ω input resistor. What is the r.m.s. noise voltage at the input to this amplifier if the ambient temperature is 27°C. [5]

b) Explain the following types of noise in details. [5]

- i) Shot noise
- ii) Partition noise

c) Write a short note on nyquist noise. [5]

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SEAT No. :

P8864

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Oct-22/TE/Insem-609

T.E. (Electronics and Computer Engineering)

DATA COMMUNICATION

(2019 Pattern) (Semester-I) (310343)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.
- 5) Use of Calculator is allowed.

Q1) a) Explain the coaxial cable in detail. **[5]**

b) Explain the following types of transmission modes in detail. **[5]**

- i) Simplex
- ii) Half duplex
- iii) Full duplex

c) Write a short note on fiber optic cables. **[5]**

OR

Q2) a) Give definitions: **[5]**

- i) Baud rate
- ii) Bit Rate
- iii) Data rate
- iv) Bandwidth

b) Explain STP & UTP in detail. **[5]**

c) Explain OSI model in detail. **[5]**

P.T.O.

- Q3)** a) Enlist the different types of internal noises and explain any two types in detail. [5]
- b) Derive the Friss's formula for noise factor of amplifiers in cascade. [5]
- c) Explain the performance of SSB-SC in presence of noise. [5]

OR

- Q4)** a) Three resistors of $10\text{ K}\Omega$, and $22\text{ K}\Omega$, and $33\text{ k}\Omega$, are at room temperature, (27°C). For a bandwidth of 100 kHz . Calculate the thermal noise voltage generated by : [5]
- i) Each resistor
- ii) Three resistors in series
- iii) Three resistors in parallel
- b) Define the terms [5]
- i) Noise figure
- ii) Noise temperature
- iii) Noise bandwidth
- iv) SNR
- c) Explain the performance of AM in presence of noise. [5]

